ScaleJade

EVALUATION REPORT

Clause Extraction from Perjanjian Baku

Evaluation of the fine-tuned model v0.1.3 against gpt-4.1-mini
Official report prepared for Bank Indonesia



October 2026

About this report

Every review of a Perjanjian Baku starts the same way. The contract is broken down into its individual clauses, each with its article number, topic and exact wording. Only then can each clause be checked against consumer-protection rules. A clause that is missed at this first step is never reviewed.

Over the past weeks, ScaleJade trained a dedicated model to do this first step, using contracts that Bank Indonesia had already broken down by hand. This report explains how the data was prepared, how the model was trained and how its accuracy was measured. It then compares the model with OpenAI's gpt-4.1-mini on 17 contracts that neither model had seen.

The model is published as `scalejade/pbbi-clause-extraction-lora`, version **v0.1.3**, which is the current stable release.

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01

Executive summary

The fine-tuned model finds more than twice as many clauses as gpt-4.1-mini, and more of what it returns is correct.

66 vs 28	**78** vs 63	**72** vs 39
clauses found per 100 in the contract	returned clauses that are correct, per 100	overall score, 0–100

Fine-tuned model (v0.1.3) vs gpt-4.1-mini, on 17 unseen contracts containing 2,074 clauses.

1. **It finds 2.3 times as many clauses.** Of every 100 clauses in Bank Indonesia's breakdown, the fine-tuned model finds 66 and gpt-4.1-mini finds 28.
2. **It is also more often correct.** 78 of every 100 clauses it returns match Bank Indonesia's version, against 63 for gpt-4.1-mini.
3. **It wins on almost every contract:** better on 13 of the 17, equal on 2.
4. **Its lead is largest on long contracts,** where it finds three times as many clauses. These are the contracts where manual review takes the most effort.
5. **It captures 97% of the contract content.** The wording of 97 of every 100 clauses in Bank Indonesia's breakdown appears in its output; only 3 are truly missing. Of the 97, 66 are split exactly as Bank Indonesia split them and 31 are split differently (Exhibit 13).
6. **It learned from Bank Indonesia's own work:** 39 contracts broken down by hand, from 41 providers and every provider-type and product category.

Recommendation. Adopt the fine-tuned model for the first step of Perjanjian Baku review, and keep improving it together through consistent labelling and more validated contracts.

The task

For each contract the model must return every clause with three things: its article number (*No. Pasal*), its topic (*Topik Klausa*) and its exact wording (*Bunyi Klausa*). Contracts run from about 15 to more than 350 clauses, and many repeat the same phrasing, which makes them easy to merge or cut short.

Because a missed clause is never reviewed, the most important measure is **how many of the clauses the model finds**. Whether what it returns is correct comes second.

02

Data

Preparing the data

Sixty contracts broken down by Bank Indonesia, sorted into categories and divided into three groups, each with one job.

Source material

The starting point is **60 Perjanjian Baku contracts from 41 providers**, covering mobile banking, internet banking, debit and credit cards, e-money, payment gateways and remittance. For each one, Bank Indonesia supplied its own breakdown in Excel with four columns: *No*, *No. Pasal*, *Topik Klausa* and *Bunyi Klausa*. Together they hold **6,890 labelled clauses**.

How each contract is categorised

Every contract is categorised from Bank Indonesia's master sheet (*PB Mainsheet*), in four layers. The same categories are stored with each contract, and all 60 were checked against the master sheet; they match.

Exhibit 1

Each contract is described by four layers of categories.

Categorisation layers

Layer	What it records	Values
1. Provider flags (*Kategori*)	Yes/no flags; a contract can carry several	Utama · BPD · Multinational · Swasta Nasional · BUMN · LSB
2. Product (*Produk*)	The service the contract covers	9 products, e.g. Mobile Banking, Kartu Debit, Uang Elektronik
3. Provider type (*Kombinasi*)	The flags combined into one label	7 types, e.g. *Utama / BUMN*, *BPD*, *Utama / LSB*
4. Category (*Tipe*)	Provider type + product	**29 categories**, e.g. *BPD / Mobile Banking*

Example: Bank Pembangunan Daerah Aceh, Mobile Banking. Flags: BPD = Y, all others T → provider type *BPD* → category *BPD / Mobile Banking*.

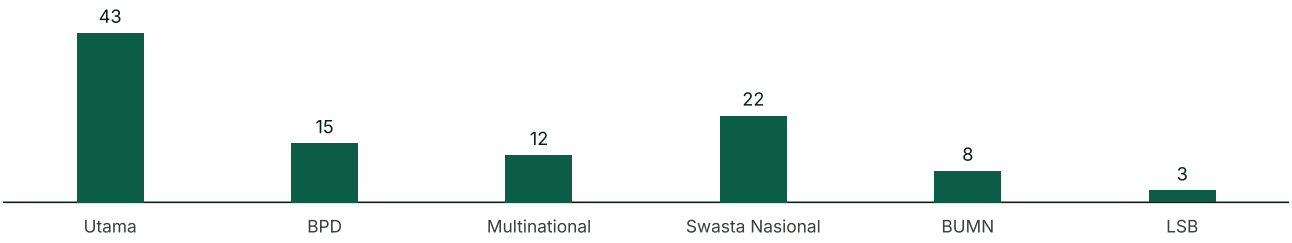
Source: Bank Indonesia PB Mainsheet; ScaleJade analysis

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Exhibit 2

Most contracts come from major (Utama) providers; non-bank e-money providers are the smallest group.

Contracts carrying each provider flag, number of contracts



A contract can carry more than one flag, so the counts add up to more than 60.

Source: Bank Indonesia PB Mainsheet; ScaleJade analysis

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Three groups, each with one job

The contracts are divided into three groups that never overlap. The model **learns** only from the training group. The validation group is used during training to **check progress** and keep the best version. The test group stays **hidden until the end** and is the basis of every result in this report.

Exhibit 3

Of the 60 contracts, 39 teach the model, 4 check its progress and 17 are kept for the final test.

Contract groups

Group	Purpose	Contracts	Clauses
Training	The model learns from these	39	4,381
Validation	Tracks progress during training and picks the best version; never learned from	4	435
Test	Final exam, kept hidden until the end; used for the comparison with gpt-4.1-mini	17	2,074
Total		**60**	**6,890**

Source: ScaleJade, October 2026

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How the categories decided the split

The split follows three rules, all based on the 29 categories:

1. **Every category is taught.** Each of the 29 categories has at least one training contract, so the model has seen every kind of contract it will meet: **29 of 29**.
2. **Test as many categories as possible. 17 categories** have contracts in both training and test, so the model is tested on kinds of contracts it learned. The other 12 have only one or

two contracts; moving one to the test group would leave nothing to learn from.

3. **The test group never changes.** The fine-tuned model and gpt-4.1-mini are measured on exactly the same 17 contracts.

Exhibit 4

A small number of contracts were moved between groups to satisfy the rules.

Moves made, and the reason for each

Contract	Moved	Why
CIMB UE Server-Based	Validation → Training	Its category had no training contract
OCBC Kartu Debit	Test → Training	Its category had no training contract
MANDIRI Kartu Debit	Test → Training	Its category had no training contract
BRI UE Chip-Based	Training → Test	So that card-based e-money is tested at all
8 contracts for v0.1.3[1]	Validation → Training	More learning material in the categories with the lowest scores; the test group was not touched

[1] ACEH Mobile Banking, BCA Internet Banking, BCA Mobile Banking, CIMB Kartu Kredit, JATENG Mobile Banking, JTRUST Mobile & Internet Banking, STANCHART Kartu Kredit, UOB Internet Banking.

Source: ScaleJade, October 2026

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Exhibit 5

Every product and provider type is represented in the training group.

Contracts per product and per provider type, by group

Product	Train.	Valid.	Test	Total
Mobile Banking	9	1	5	**15**
Kartu Debit	8	2	3	**13**
Internet Banking	7	0	2	**9**
Kartu Kredit	6	0	2	**8**
UE Server-Based	2	1	2	**5**
Mobile & Internet Banking	3	0	1	**4**
UE Chip-Based	2	0	1	**3**
Payment Gateway	1	0	1	**2**
Remitansi	1	0	0	**1**

Provider type	Train.	Valid.	Test	Total
Utama / Swasta Nasional	9	1	5	**15**
Utama / Multinational	8	0	4	**12**
BPD	7	1	2	**10**
Utama / BUMN	5	1	2	**8**
Swasta Nasional	6	0	1	**7**
Utama / BPD	3	0	2	**5**
Utama / LSB	1	1	1	**3**

Source: Bank Indonesia PB Mainsheet; ScaleJade analysis

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Exhibit 6

All 29 categories have a training contract; 17 are also tested.

Coverage map, number of contracts per provider type × product, by group

	Mobile Banking	Kartu Debit	Internet Banking	Kartu Kredit	UE Server-Based	Mobile & Internet Banking	UE Chip-Based	Payment Gateway	Remitansi
Utama / Swasta Nasional	2 1	1 1 1	2 1	2 1	1 1		1		
Utama / Multinational	1 1	1	2 1	2 1		2 1			
BPD	2 1	3 1 1	1			1			
Utama / BUMN	1 1 1	1	1	1			1 1		
Swasta Nasional	1	1	1	1				1 1	1
Utama / BPD	2 1	1 1							
Utama / LSB					1 1 1				

Training Validation Test

Each cell shows the number of contracts in training, validation and test for that category. Empty cells are combinations with no contract.

Source: Bank Indonesia PB Mainsheet; ScaleJade analysis

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How each contract was prepared

- **Same text as in real use.** Each PDF is read with the same text pipeline the production application uses, so the model learns from exactly what it will see in practice.
- **One whole contract is one example.** Contracts are never cut into pieces, so the model learns to handle full documents, from about 15,000 to 89,000 tokens (word pieces) long.
- **Bank Indonesia's breakdown is the answer key,** written in the output format the application expects.

03

Training the model

We did not build a language model from scratch. We started from one that already reads Indonesian, and taught it this one task by example.

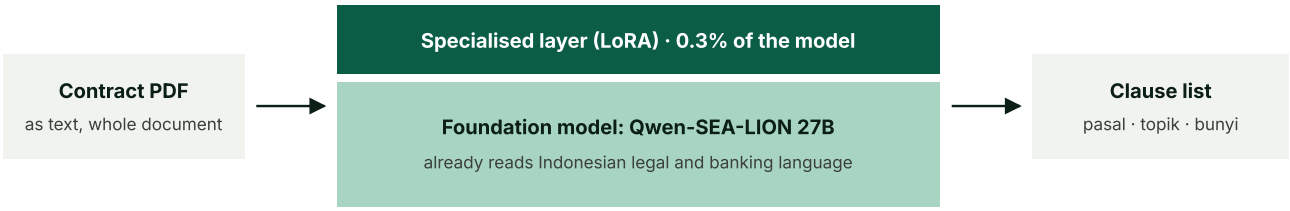
What the model is

The model is published as `scalejade/pbbi-clause-extraction-lora`, version **v0.1.3**, tagged as the stable release. It has two parts. The **foundation** is Qwen-SEA-LION 27B, an open model built for Southeast Asian languages that already understands Indonesian legal and banking language. On top of it sits a **small specialised layer** (a LoRA adapter), only 0.3% of the model's size, trained on Bank Indonesia's breakdowns. The foundation's general knowledge stays intact, and the specialised layer can be improved or replaced on its own.

Exhibit 7

A small specialised layer on top of an Indonesian-language foundation model.

How the model is built



Source: ScaleJade, October 2026

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Exhibit 8

Training follows five steps, from foundation to final test.

Training process

1	2	3	4	5
Foundation	**Learning material**	**Teach by example**	**Check and keep the best**	**Final test**
Qwen-SEA-LION 27B, an open model built for Southeast Asian languages, including Indonesian	60 contracts with Bank Indonesia's breakdown, read exactly as the application reads them	39 contracts; a small specialised layer learns Bank Indonesia's way of splitting clauses	4 validation contracts track progress; only the best version is kept	17 hidden test contracts, compared with gpt-4.1-mini

Source: ScaleJade, October 2026

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Checking progress and keeping the best version

The model reads each training contract together with Bank Indonesia's breakdown and adjusts itself step by step. Every 4 steps it is scored on the validation contracts, which it never learns from. The score is an error measure, so lower is better. When it stops falling and starts to rise, the model has begun memorising its training contracts instead of learning the method, and only the best version is kept.

Exhibit 9

The best version came after about one full read of the training contracts.

Error on the validation contracts during training, lower is better



The model was scored every 4 steps on the 4 validation contracts. Error fell until step 20 and then flattened, so the step-20 version was kept and released as v0.1.3.

Source: ScaleJade training logs, October 2026

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04

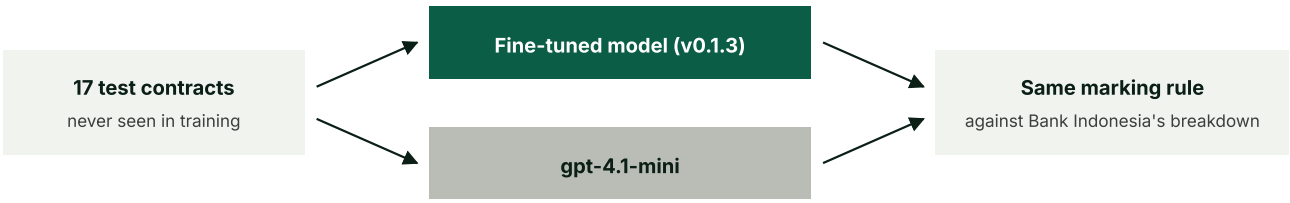
Measuring accuracy

Both models were given the same contracts and the same request, and every answer was marked with the same rule against Bank Indonesia’s breakdown.

Exhibit 10

Both models received identical input and identical marking.

How the test was run



Both models were set to give their single most consistent answer. gpt-4.1-mini is also reported at its default setting, where it scores higher (47), so the result does not depend on this choice.

Source: ScaleJade, October 2026

ScaleJade

When does a clause count as found?

Each clause a model returns is compared with Bank Indonesia's breakdown. **It counts as a match when it shares at least 80% of its words with one of Bank Indonesia's clauses.** Matching is one to one: each of Bank Indonesia's clauses can be matched only once, closest pairs first, so returning the same clause twice does not earn two matches.

The rule is strict about splitting. A clause whose words are all present but which is merged with another, or cut in half, counts as a miss.

Exhibit 11

A clause matches only when its wording and its boundaries agree with Bank Indonesia’s breakdown.

Illustrative example of the matching rule

Clause in Bank Indonesia's breakdown

Nasabah wajib menjaga kerahasiaan PIN dan tidak memberitahukan PIN kepada pihak mana pun, termasuk petugas Bank.

MATCH

Same clause, punctuation differs

Nasabah wajib menjaga kerahasiaan PIN, dan tidak memberitahukan PIN kepada pihak mana pun termasuk petugas Bank.

Almost all words shared: well above 80%

NO MATCH

Two clauses merged into one

Nasabah wajib menjaga kerahasiaan PIN dan tidak memberitahukan PIN kepada pihak mana pun, termasuk petugas Bank. Segala kerugian akibat kelalaian Nasabah dalam menjaga PIN menjadi tanggung jawab Nasabah sepenuhnya.

The extra sentence pulls the overlap below 80%

NO MATCH

One clause split in half

Nasabah wajib menjaga kerahasiaan PIN.

Covers only part of the clause: under 80%

The clause shown is written for illustration and is not quoted from any contract.

Source: ScaleJade

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Recall, precision and F1

Three measures, in plain words

Recall · “Did it find everything?”

Of all the clauses in Bank Indonesia's breakdown, how many did the model find? This measures **what was missed**, and matters most for compliance review.

Precision · “Is what it returned correct?”

Of all the clauses the model returned, how many match a real clause? This measures **what was wrong or extra**.

F1 · “One overall score”

Combines recall and precision into one number, shown here as 0–100. It is pulled toward the weaker of the two, so a model cannot score well by being strong on one and poor on the other.

Why both are needed: a model that returns every sentence finds everything but is mostly wrong; a model that returns only ten clauses is right but misses most. F1 is high only when both are good.

Exhibit 12

The fine-tuned model scores higher on both parts of the overall score.

Worked example with the real totals, 17 test contracts

	Fine-tuned model (v0.1.3)	gpt-4.1-mini
Clauses in Bank Indonesia's breakdown	2,074	2,074
Clauses the model returned	1,741	929
Matched (found)	1,365	587
Recall = matched ÷ clauses in the breakdown	1,365 ÷ 2,074 = **0.66**	587 ÷ 2,074 = 0.28
Precision = matched ÷ clauses returned	1,365 ÷ 1,741 = **0.78**	587 ÷ 929 = 0.63
F1 = 2 × P × R ÷ (P + R)	**0.72**	0.39

Source: ScaleJade evaluation, October 2026

ScaleJade

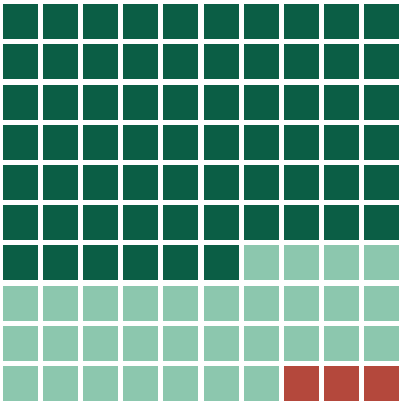
A second view: how much of the contract is captured

The strict match asks whether a clause is split *exactly* as Bank Indonesia split it. A second check asks a simpler question: **does the clause's wording appear anywhere in the model's output?** A clause counts as captured when at least 60% of its words appear in the output, and as missing when fewer do.

Exhibit 13

The fine-tuned model captures the wording of 97 of every 100 clauses; only 3 are truly missing.

What happens to every 100 clauses in Bank Indonesia's breakdown, fine-tuned model v0.1.3, 17 test contracts



97 **of every 100 clauses**
have their wording in the model's output

66 **Found, and split exactly as Bank Indonesia split it**
1,365 clauses · counted in the strict score

31 **Present, but split differently**
649 clauses · e.g. two clauses kept together, or one divided in two

3 **Truly missing**
60 clauses · fewer than 60% of their words anywhere in the output

Each square is one clause in 100. Totals over 2,074 labelled clauses: 1,365 found exactly, 649 present but split differently, 60 missing.

Source: ScaleJade evaluation, October 2026

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The gap between 66 and 100 is therefore mostly about **where a clause begins and ends**, not lost content, which is why consistent labelling is the most important next step.

05

Results

On every summary measure the fine-tuned model is ahead, and its lead is largest where review is hardest.

Exhibit 14

The fine-tuned model is ahead on every summary measure.

Overall results, 17 unseen contracts, 2,074 clauses

Measure	Fine-tuned model (v0.1.3)	gpt-4.1-mini
Clauses found, per 100 in the contract (recall)	**66**	28
Returned clauses that are correct, per 100 (precision)	**78**	63
Overall score, 0–100 (F1)	**72**	39
Contracts with the higher score	**13** (2 equal)	2
Clauses returned, relative to the contract	**84%**	45%

Source: ScaleJade evaluation, October 2026

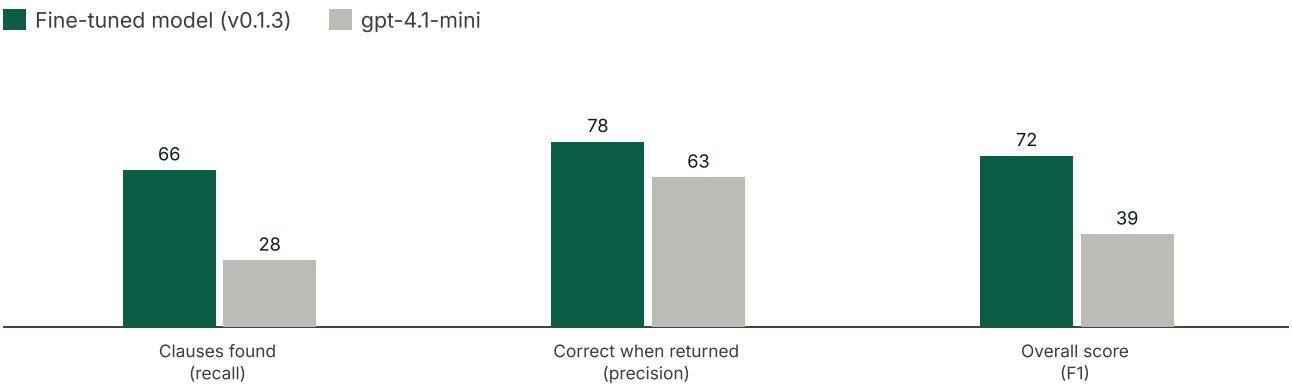
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Evidence 1: more than twice as many clauses found

Exhibit 15

The fine-tuned model finds 2.3 times as many clauses, and more of what it returns is correct.

Recall, precision and overall score, 0–100



Source: ScaleJade evaluation, October 2026

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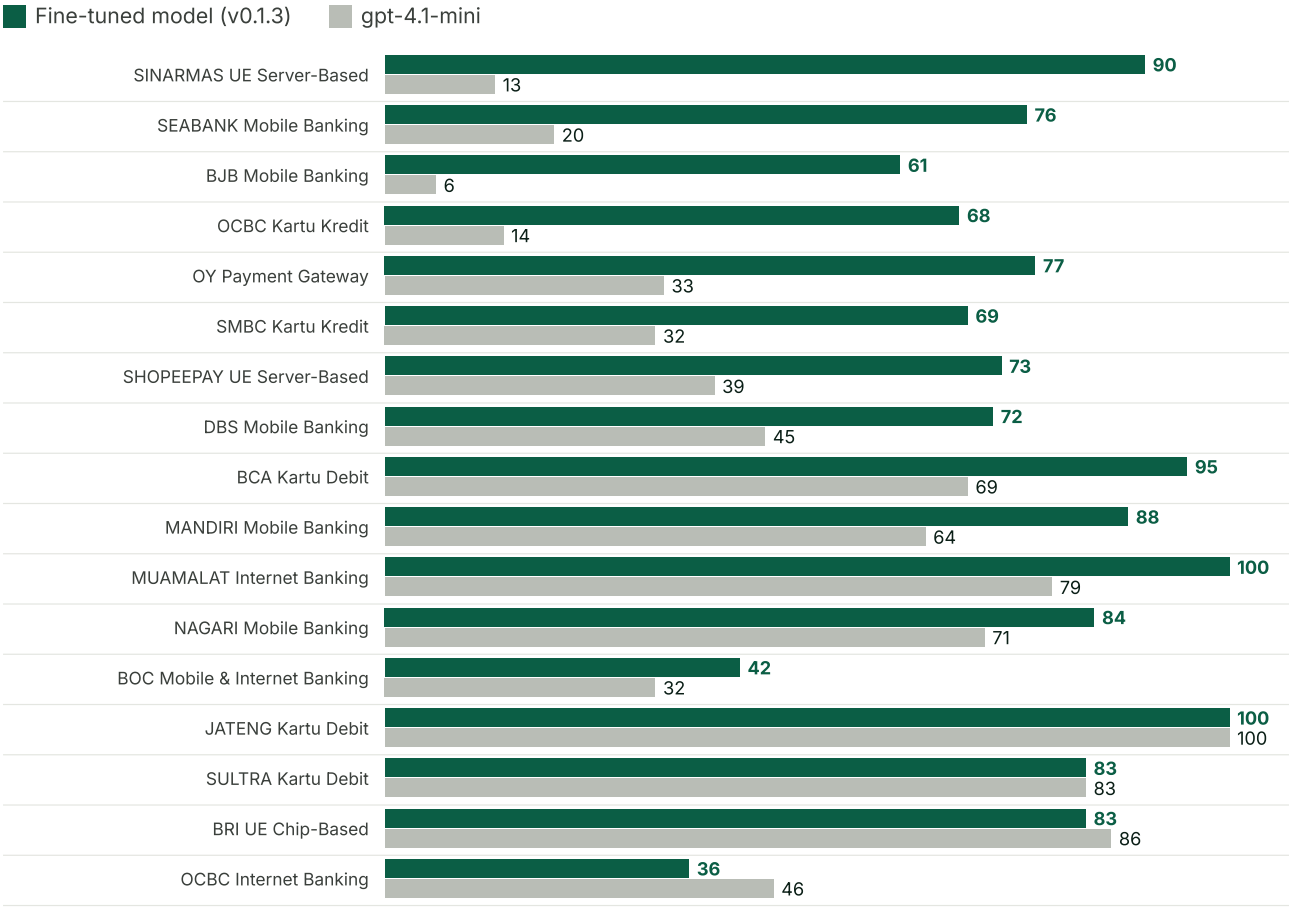
Evidence 2: it wins on almost every contract

The fine-tuned model has the higher score on 13 of the 17 contracts and the same score on 2. gpt-4.1-mini is ahead on 2.

Exhibit 16

The fine-tuned model scores higher on 13 of 17 contracts.

Overall score per contract, 0–100, sorted by the size of the lead



Source: ScaleJade evaluation, October 2026

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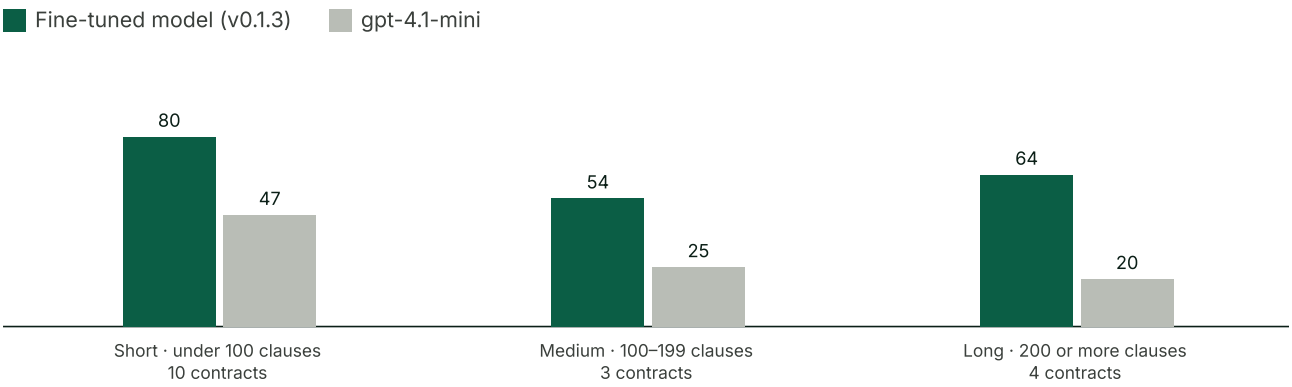
Evidence 3: its largest lead is on long contracts

On short contracts both models do reasonably well. As contracts get longer, gpt-4.1-mini increasingly merges clauses or stops part-way through. The fine-tuned model keeps Bank Indonesia's level of detail.

Exhibit 17

On contracts with 200 or more clauses, the fine-tuned model finds three times as many.

Clauses found per 100, by contract length



Source: ScaleJade evaluation, October 2026

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On the longest contracts, the general-purpose model finds 20 of every 100 clauses. The fine-tuned model finds 64.

Evidence 4: gpt-4.1-mini stops early

On several contracts gpt-4.1-mini returns only a small fraction of the clauses. In every case it ended its answer on its own, well within its output limit; it was not cut off. Overall it returns 45% as many clauses as Bank Indonesia's breakdown contains, against 84% for the fine-tuned model.

Exhibit 18

On some contracts gpt-4.1-mini returns only a handful of clauses.

Clauses found on four contracts

Contract	Clauses in the contract	Found by gpt-4.1-mini	Found by the fine-tuned model
BJB Mobile Banking	110	4	**69**
SINARMAS UE Server-Based	49	4	**44**
OCBC Kartu Kredit	233	19	**131**
SEABANK Mobile Banking	358	42	**247**

Source: ScaleJade evaluation, October 2026

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Evidence 5: consistent across categories

Using the same categories as in Chapter 2, the fine-tuned model scores higher for 6 of the 8 products tested and for all 7 provider types.

Exhibit 19

The lead holds across products and provider types.

Overall score by product and by provider type, 0–100

Product	Contracts	Fine-tuned	gpt-4.1-mini
Kartu Debit	3	**94**	76
UE Chip-Based	1	83	**86**
Payment Gateway	1	**77**	33
UE Server-Based	2	**76**	34
Mobile Banking	5	**75**	36
Kartu Kredit	2	**68**	23
Internet Banking	2	51	**53**
Mobile & Internet Banking	1	**42**	32

Provider type	Contracts	Fine-tuned	gpt-4.1-mini
Utama / BUMN	2	**87**	70
BPD	2	**84**	75
Utama / Swasta Nasional	5	**79**	35
Swasta Nasional	1	**77**	33
Utama / LSB	1	**73**	39
Utama / BPD	2	**66**	24
Utama / Multinational	4	**59**	36

Source: ScaleJade evaluation, October 2026

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Where gpt-4.1-mini is ahead

gpt-4.1-mini scores higher on two contracts: **OCBC Internet Banking** (46 against 36) and **BRI card-based e-money** (86 against 83, a small difference). The OCBC result traces to how lists of definitions are labelled. In some training contracts several definitions are grouped into one clause; in others each definition is its own clause. Because the examples point both ways, the model cannot learn one rule. This can be fixed together by agreeing one labelling rule (see Chapter 6).

06

Outlook

Why a dedicated model suits Bank Indonesia, what the test cannot show, and the next steps to take together.

Why a dedicated model suits Bank Indonesia

- **Contract data stays on controlled infrastructure.** The model runs on infrastructure we control; contracts are not sent to a third-party AI service.
- **It follows Bank Indonesia's own method.** It learned from Bank Indonesia's breakdowns, so its output follows the same conventions.
- **Every new version is measured before release.** Each version is re-tested on the same 17 contracts, so any change is measured, not assumed.
- **The trade-off is speed.** It is slower per contract: about 16 minutes for the longest test contract, against under 3 minutes for gpt-4.1-mini. For review work that is not time-critical, completeness matters more.

Limitations

- **Sample size.** The test covers 17 contracts. Several products were tested on only one or two, so results per product or provider type are indicative.
- **Strict marking.** A clause split slightly differently from Bank Indonesia's version counts as a miss even when all its words are present, so the model's practical usefulness is somewhat higher than its score.
- **No tailoring of gpt-4.1-mini.** It received the same instructions as the fine-tuned model, with no examples or extra training.

Next steps together

1. **Deploy v0.1.3 in the application**
   ScaleJade
2. **Agree one labelling rule for definitions** and lists of sub-clauses, and apply it to existing contracts
   Bank Indonesia with ScaleJade
3. **Label more contracts** for the next version. The current timeline did not allow more labelling; more labelled contracts will train a more accurate extraction model
   Bank Indonesia with ScaleJade

A model trained on Bank Indonesia's own work finds more than twice as many clauses as gpt-4.1-mini. With more labelled contracts, the next version can close the remaining gap further.

Appendix A · Results per contract

Both models set to their most consistent answer. Score = F1 × 100. The higher score in each row is highlighted.

Contract	Clauses	Fine-tuned: found	Fine-tuned: score	gpt-4.1-mini: found	gpt-4.1-mini: score
JATENG Kartu Debit	15	15	100	15	100
MUAMALAT Internet Banking	46	46	**100**	32	79
BCA Kartu Debit	87	79	**95**	50	69
SINARMAS UE Server-Based	49	44	**90**	4	13
MANDIRI Mobile Banking	98	88	**88**	53	64
NAGARI Mobile Banking	44	37	**84**	27	71
BRI UE Chip-Based	27	25	83	25	**86**
SULTRA Kartu Debit	18	15	83	15	83
OY Payment Gateway	89	67	**77**	20	33
SEABANK Mobile Banking	358	247	**76**	42	20
SHOPEEPAY UE Server-Based	211	130	**73**	57	39
DBS Mobile Banking	262	168	**72**	95	45
SMBC Kartu Kredit	172	127	**69**	45	32
OCBC Kartu Kredit	233	131	**68**	19	14
BJB Mobile Banking	110	69	**61**	4	6
BOC Mobile & Internet Banking	79	25	**42**	18	32
OCBC Internet Banking	176	52	36	66	**46**
All 17	**2,074**	**1,365**	**72**	**587**	**39**

Appendix B · The 60 contracts

Every contract with its product, provider type and group. Categories from Bank Indonesia's PB Mainsheet.

Provider	Product	Provider type	Group
Bank Pembangunan Daerah Aceh	Mobile Banking	BPD	Training
Bank Pembangunan Daerah Banten	Kartu Debit	BPD	Training
Bank Pembangunan Daerah Banten	Mobile Banking	BPD	Training
PT Bank Central Asia Tbk	Internet Banking	Utama / Swasta Nasional	Training
PT Bank Central Asia Tbk	Mobile Banking	Utama / Swasta Nasional	Training
PT Bank Negara Indonesia Tbk	Kartu Kredit	Utama / BUMN	Training
PT Bank Syariah Indonesia Tbk	Mobile Banking	Utama / BUMN	Training
PT Bank Tabungan Negara Tbk	Internet Banking	Utama / BUMN	Training
PT Bank CIMB Niaga Tbk	Kartu Kredit	Utama / Swasta Nasional	Training
PT Bank CIMB Niaga Tbk	UE Server-Based	Utama / Swasta Nasional	Training
PT Bank Danamon Indonesia Tbk	UE Chip-Based	Utama / Swasta Nasional	Training
PT Dompet Anak Bangsa	UE Server-Based	Utama / LSB	Training
PT Honest Financial Technologies	Kartu Kredit	Swasta Nasional	Training
PT Bank Jago Tbk	Mobile Banking	Swasta Nasional	Training
Bank Pembangunan Daerah Jawa Tengah	Mobile Banking	Utama / BPD	Training
Bank Pembangunan Daerah Jawa Timur	Kartu Debit	Utama / BPD	Training
Bank Pembangunan Daerah Jawa Timur	Mobile Banking	Utama / BPD	Training
PT Bank JTrust Indonesia Tbk	Mobile & Internet Banking	Utama / Multinational	Training
Bank Pembangunan Daerah Kalimantan Selatan	Kartu Debit	BPD	Training
Bank Pembangunan Daerah Kalimantan Timur dan Utara	Mobile & Internet Banking	BPD	Training
PT Bank Mandiri Tbk	Kartu Debit	Utama / BUMN	Training
PT Bank Mandiri Tbk	UE Chip-Based	Utama / BUMN	Training
PT Bank Maspion Indonesia Tbk	Internet Banking	Swasta Nasional	Training
PT Bank Maspion Indonesia Tbk	Kartu Debit	Swasta Nasional	Training
PT Bank Maybank Indonesia Tbk	Mobile & Internet Banking	Utama / Multinational	Training
PT Bank Mega Tbk	Kartu Kredit	Utama / Swasta Nasional	Training
PT Bank Mega Tbk	Mobile Banking	Utama / Swasta Nasional	Training
Bank Pembangunan Daerah Nusa Tenggara Timur	Internet Banking	BPD	Training
PT Bank OCBC NISP Tbk	Kartu Debit	Utama / Multinational	Training
PT Bank Sinarmas Tbk	Kartu Debit	Utama / Swasta Nasional	Training
PT Bank SMBC Indonesia Tbk	Internet Banking	Utama / Swasta Nasional	Training
Standard Chartered Bank	Internet Banking	Utama / Multinational	Training
Standard Chartered Bank	Kartu Kredit	Utama / Multinational	Training

Provider	Product	Provider type	Group
Bank Pembangunan Daerah Sulawesi Selatan dan Barat	Kartu Debit	BPD	Training
PT Bank UOB Indonesia	Internet Banking	Utama / Multinational	Training
PT Bank UOB Indonesia	Kartu Kredit	Utama / Multinational	Training
PT Bank UOB Indonesia	Mobile Banking	Utama / Multinational	Training
WISE	Remitansi	Swasta Nasional	Training
PT Yukk Kreasi Indonesia	Payment Gateway	Swasta Nasional	Training
PT Bank Negara Indonesia Tbk	Mobile Banking	Utama / BUMN	Validation
PT Bank CIMB Niaga Tbk	Kartu Debit	Utama / Swasta Nasional	Validation
Bank Pembangunan Daerah Jambi	Kartu Debit	BPD	Validation
PT Visionet International	UE Server-Based	Utama / LSB	Validation
PT Bank Central Asia Tbk	Kartu Debit	Utama / Swasta Nasional	Test
Bank Pembangunan Daerah Jawa Barat dan Banten Tbk	Mobile Banking	Utama / BPD	Test
Bank of China Ltd Cabang Jakarta	Mobile & Internet Banking	Utama / Multinational	Test
PT Bank Rakyat Indonesia Tbk	UE Chip-Based	Utama / BUMN	Test
PT Bank DBS Indonesia	Mobile Banking	Utama / Multinational	Test
Bank Pembangunan Daerah Jawa Tengah	Kartu Debit	Utama / BPD	Test
PT Bank Mandiri Tbk	Mobile Banking	Utama / BUMN	Test
PT Bank Muamalat Indonesia Tbk	Internet Banking	Utama / Swasta Nasional	Test
Bank Pembangunan Daerah Sumatera Barat	Mobile Banking	BPD	Test
PT Bank OCBC NISP Tbk	Internet Banking	Utama / Multinational	Test
PT Bank OCBC NISP Tbk	Kartu Kredit	Utama / Multinational	Test
PT Dompet Harapan Bangsa	Payment Gateway	Swasta Nasional	Test
PT Bank Seabank Indonesia	Mobile Banking	Utama / Swasta Nasional	Test
PT Airpay International Indonesia	UE Server-Based	Utama / LSB	Test
PT Bank Sinarmas Tbk	UE Server-Based	Utama / Swasta Nasional	Test
PT Bank SMBC Indonesia Tbk	Kartu Kredit	Utama / Swasta Nasional	Test
Bank Pembangunan Daerah Sulawesi Tenggara	Kartu Debit	BPD	Test